# WASP-35b

R-1484

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-35_191031 · created 2026-08-01T13:13:33+00:00

**PRACTICE / VALIDATION RUN — not submitted to any science database**

## Results

| | |
|---|---|
| Mid-Transit Time (Tmid) | 2458787.898 +/- 0.0021 BJD_TDB |
| Ratio of Planet to Stellar Radius (Rp/R*) | 0.165 +/- 0.048 |
| Transit depth (Rp/Rs)^2 | 2.72 +/- 1.57 [%] |
| Orbital Inclination (inc) | 88.98 +/- 2.05 |
| Airmass coefficient 1 (a1) | 0.41 +/- 0.24 |
| Airmass coefficient 2 (a2) | 0.55 +/- 0.56 |
| Scatter in the residuals of the lightcurve fit is | 52.68 % |
| Best Comparison Star | None |
| Optimal Aperture | 0.75 |
| Optimal Annulus | 3.0 |
| Transit Duration (day) | 0.135 +/- 0.011 |

## Accuracy vs published ephemeris (ExoClock/NEA)

| | |
|---|---|
| Rp/R■ fit vs published | 0.165 ± 0.048 vs 0.1241 (deviation 0.83σ) |
| Duration fit vs published | 0.135 d vs 0.1317 d (deviation 0.29σ) |
| Mid-time O–C | 3.7 min |
| Depth SNR | 3.3 |
| Residual scatter | 52.68 % |

## Light curve

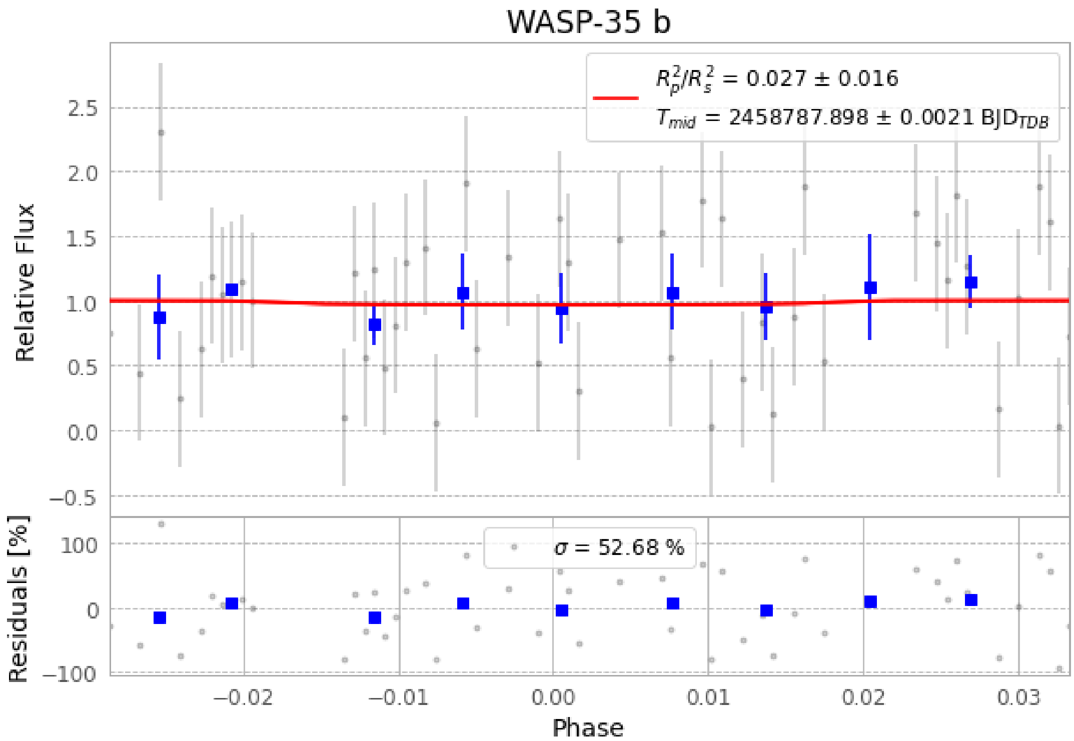


Figure 1 — FinalLightCurve_WASP-35 b_2019-10-31.png (SHA-256 07acbde644f9f6b0…)

## Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [65, 245], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 9edcb3815c6ca12f…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

## Data provenance

95 FITS frames (62 MB), `WASP-351910031071512.FITS` → `WASP-351910031115712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-35-191031.log (6ff84c907ffb…), FinalLightCurve_WASP-35 b_2019-10-31.png (07acbde644f9…), FinalLightCurve_WASP-35 b_2019-10-31.csv (d84291c68cf3…)

## Compute resources

Wall time 165.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)